Sidharth Bhagat

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Graduate Software Engineer / AMTS
Electrical engineering graduate candidate with experience across software engineering, data platforms and applied AI. Builds dependable backend workflows and accessible interfaces, with strong algorithms and competitive-programming foundations.

EDUCATION

Indian Institute of Technology Jodhpur 2023–2027
B.Tech, Electrical Engineering | CGPA: 7.88/10 | Departmental rank 3 at end of first year.

EXPERIENCE

Software Engineer Intern – Data Engineering | ICICI Bank, Audit Intelligence Unit Summer 2026

- Built AIU DataVerse, a Python platform for heterogeneous financial data, with 9-phase Bayesian delimiter detection and 23-stage type recovery for automated schema inference.
- Migrated Pandas processing to Polars LazyFrame with SHA-256 content-addressed Parquet caching; measured 1,673x profiling and 12.91x CSV-parsing speedups on internship benchmark workloads.
- Implemented regression modeling, MinHash/LSH deduplication and population stability index (PSI) drift detection; audited the codebase and remediated 6 defects.

Software Development Fellow | SoLA, Comic Studies Summer 2025

- Built a comic-to-audio pipeline integrating computer vision, OCR, dialogue extraction and speech synthesis; processed 4 pages/minute in the fellowship workflow.
- Implemented speaker attribution for up to 5 characters and emotion-conditioned narration, preserving dialogue identity through the audio pipeline.

Undergraduate Research Intern | IIT Jodhpur, CS Department Summer 2024

- Benchmarked meta-heuristic and approximation approaches to combinatorial optimization with Prof. Tanmay Nitin Inamdar; achieved a 40% improvement in solution quality in the studied scenarios.

PROJECTS

WARDEN | Reliable backend systems

- Built authenticated FastAPI services with PostgreSQL, Redis Streams and transactional outboxes; recovered 500/500 jobs across 12 worker kills with no lost jobs or duplicate durable effects in crash tests.
- Implemented a typed rule language with a Pratt parser and RE2, exercised by 60,000 fuzz cases; added signed webhooks, retries, circuit breakers and dead-letter queues.
- Tuned PostgreSQL queries with indexes and EXPLAIN ANALYZE, reducing execution time by up to 99.6% on 150,002 fixture rows; added OpenTelemetry, Prometheus and Grafana observability.

Panel | Full-stack accessible AI

- Built a React/TypeScript reader and FastAPI/SQLite backend combining PDF extraction, EasyOCR and OpenCLIP embeddings with editable character casting and cross-page identity merging.
- Integrated Kokoro/ONNX speech with eSpeak fallback, cached audio, 28 English voice definitions and 13 prosody styles; added transcript controls, reading progress and accessibility preferences.

CIFAR-10 | Machine learning experiments

- Compared classical classifiers, clustering methods and neural networks across 13 experiment notebooks; achieved 80.40% CIFAR-10 test accuracy.
- Built a Keras convolution/pooling/dropout model and Flask demonstration for training, model persistence and image prediction; evaluated preprocessing and model predictions.

TECHNICAL SKILLS

Programming: Python, C++, SQL, JavaScript, TypeScript; algorithms, data structures, OOP, concurrency.

Systems and data: FastAPI, Django REST Framework, PostgreSQL, Redis, Polars, Pandas, Parquet.

Applied AI and delivery: TensorFlow, Keras, scikit-learn, OpenCV, React, Docker, Git, Linux, observability.

ACHIEVEMENTS AND LEADERSHIP

- LeetCode Guardian (top 1.02%); CodeChef 5-star; 800+ problems solved.
- Top 1.5% among 1.2M+ JEE candidates; Assistant Head, Varchas 2025 and Prometeo 2025, IIT Jodhpur.